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The **POST (Power-On Self-Test)** process is a diagnostic testing sequence run automatically by a computer's firmware (BIOS or UEFI) **when the system is powered on**. It checks that the hardware components are working properly before loading the operating system.

🔁 Steps in the POST Process:

1. **Power Supply Check:**
   - The **SMPS (Power Supply)** sends a "Power Good" signal to the motherboard if the voltage levels are stable.

2. **CPU Initialization:**
   - The **processor starts executing** code from a predefined location in the BIOS ROM.
3. **BIOS/UEFI Execution:**
   - BIOS/UEFI begins the POST routine.
4. **Memory Check (RAM):**
   - BIOS tests the system's **RAM** to ensure it is functioning correctly.
5. **Hardware Detection:**
   - It checks for **keyboard, mouse, display, hard disk, optical drive,** etc.
6. **Video Card Initialization:**
   - BIOS initializes the **graphics card** to display messages on screen.
7. **Beep Codes (if errors are found):**
   - If there's no display, the **motherboard speaker** gives specific **beep sounds** to indicate hardware issues (e.g., RAM failure, GPU issue).
8. **Display POST Screen:**
   - If successful, it shows details like CPU type, RAM, connected drives, etc.
9. **Boot Device Detection:**
   - BIOS identifies the device (HDD/SSD, USB, CD/DVD) from which to **load the OS**.
10. **Hand Over to Bootloader:**
    - Once POST is successful, control is passed to the **bootloader** to start the operating system.

🔥 What is Booting?

Booting is the process of **starting the computer** and **loading the operating system** into RAM so that it can be used.

⚙️ **Steps of the Boot Process:**

1. Power ON

- When you press the power button:
  - The **power supply (SMPS)** sends signals to the motherboard and other components.
  - The **CPU** starts working and looks for firmware instructions.

2. POST (Power-On Self-Test)

- Run by **BIOS/UEFI** (firmware stored in ROM).
- Checks hardware components: RAM, keyboard, GPU, etc.
- If there's a problem, you may hear **beep codes** or error messages.

3. BIOS/UEFI Execution

- After successful POST, BIOS/UEFI:
  - Detects connected **storage devices** (HDD/SSD, USB, etc.).
  - Searches for a **bootable device**.

4. Bootloader Execution

- BIOS/UEFI finds and loads the **Bootloader** (like **GRUB** for Linux or **Windows Boot Manager**).
- The bootloader is stored in:
  - **MBR (Master Boot Record)** – for MBR-partitioned disks.
  - **EFI System Partition** – for GPT/UEFI systems.

5. Loading OS Kernel

- Bootloader loads the **OS Kernel** (core of the OS) from the selected partition (usually C: drive).
- The kernel is responsible for:
  - Initializing memory and CPU
  - Starting drivers and hardware interfaces

6. System Initialization

- OS kernel starts **system processes** (like init, wininit.exe, systemd, etc.).
- Loads:
  - **Device drivers**
  - **Login screen**
  - **Services** (like network, audio, etc.)

7. User Login

- Login screen appears (GUI or CLI).
- After login, the system loads **user profile**, **desktop**, and **startup applications**.

💡 Boot Sequence Summary (Windows/Linux)

Step	Description
Power On	SMPS powers motherboard
POST	BIOS tests hardware
BIOS/UEFI	Locates boot device
Bootloader	GRUB/Windows Boot Manager
Load Kernel	OS core is loaded into RAM
Init System	Drivers and services start
Login	User interface appears